

Compiling Typed APIs into Context-Free Languages

Breandan Considine, Clemente Pasti, Tim Vieira

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Motivation: type-safe code completion

- ▶ Construct a $\ell_\Gamma = \mathcal{L}(G_\Gamma)$ consisting of well-typed statements
- ▶ No well-typed statement $\sigma \notin \ell_\Gamma$ or ill-typed statements $\sigma \in \ell_\Gamma$
- ▶ For any $\sigma \in \Sigma^*$ such that $\exists \tau. \sigma\tau \in \ell_\Gamma$, show τ 's in shortlex order

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```
1 #include <memory>
2 long* codes(); double* groups();
3 const char* text(); float* samples();
4 std::unique_ptr<int> fresh(), parse(const char*);
5 std::unique_ptr<int> average(float*), select(double&);
6 std::unique_ptr<int> combine(const char*, long*);
7 std::unique_ptr<int> schedule(const char*, long*, float*);
8
9 int main(int argc, char** args) {
10     std::unique_ptr<int> b
11 }
```

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9 int main(int argc, char** args) {
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```

```
;
= fresh();
= average(samples());
= parse(text());
= select(groups()[argc]);
= combine(text(), codes());
= schedule(text(), codes(), samples());
```

Existing work on type-safe code completion

Legend.  by construction **gen** generation-time **filt** post-generation  partial / empirical **—** absent / not claimed

Lens	Type-directed search		Spec-driven	Learned completion		Constrained	decoding
Feature	Perelman'12	Gvero'13	Pelsmaeker'22	Raychev'14	Semenkin'24	Melcer'24	Mündler'25
Language	C#	Scala	Statix DSL	Java	Python+JVM+JB	Python 3	TypeScript
Output unit	expression	expression	fragment	call sequence	full line	FIM span	function
Query form	partial expr	typed hole	caret	holes	left prefix	FIM context	prompt
Candidate set	type graph	typed terms	Statix terms	API traces	LLM output	parser states	typed tokens
Ranking signal	type distance	usage priors	solver order	n-gram/RNN	local LM	host LM	host LM
Fill model	hole	hole	cursor	cursor	cursor	FIM	prefix
Learned signal	—		—				
Filter timing	search	search	solver	rank only	filt	gen	gen
Syntax safe					filt	gen	gen
Type safe					—	—	gen
Statement-scale	—	—					
Parametricity	—	—		—	—		
Candidate complete				—	—	—	—
Usable tool?	VS proto	Eclipse	Spoofax	prototype	IntelliJ Platform plugin	parser	Git repo
Public code/tool	—	code	artifact	—	plugin	code	code

Gap. Prior work tends to trade off probabilistic ranking, statement-scale completion, and formal type guarantees but does not enumerate a ranked language of complete typed statements.

[1] Perelman et al., PLDI'12

[2] Gvero et al., PLDI'13 / code

[3] Pelsmaeker et al., OOPSLA'22 / artifact

[4] Raychev et al., PLDI'14 / BigCode

[5] Semenkin et al., arXiv:2405.08704 / plugin

[6] Melcer et al., arXiv:2402.17988 / code

[7] Mündler et al., PLDI'25 / code

How long are typical source code statements?

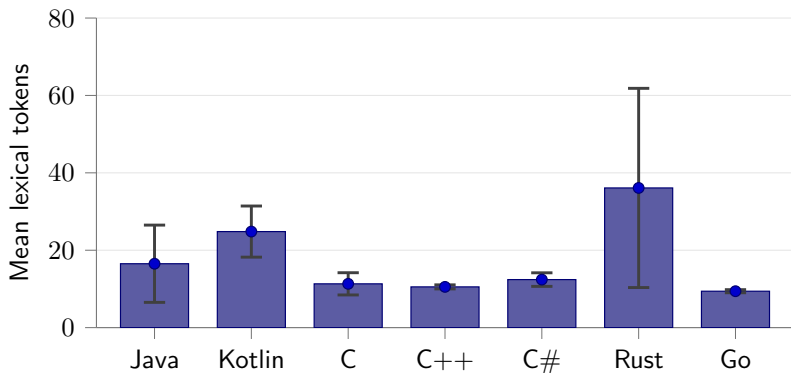


Figure: Whiskers are 95% CIs over sampled files. Statements are Tree-sitter statement nodes with comments and whitespace ignored. Samples are random GitHub files > 20kb; per-language sample sizes are shown below.

How large are programming language grammars?

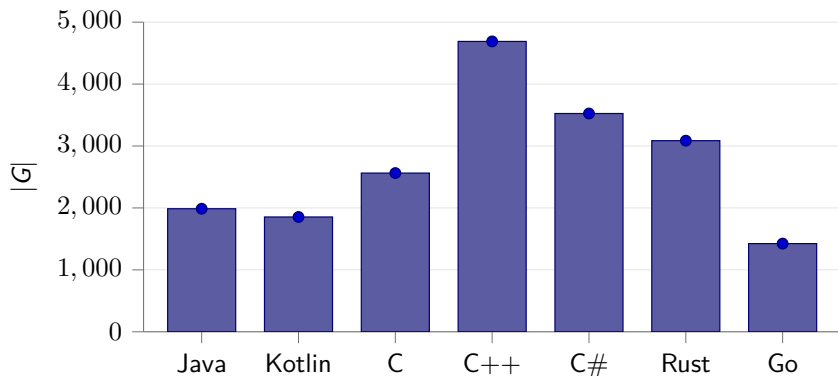


Figure: Grammar size is computed from the Tree-sitter reference CFGs, where $|G|$ is defined as $\sum_{w \in V} \sum_{(w \rightarrow \sigma) \in P} |w\sigma|$.

Correctness criterion: what must the CFG preserve?

Surface fragment G'

e.g., names, invocation, member chains

$S ::= E; \quad E ::= \text{ID} \mid \text{ID}() \mid \text{ID}(\text{ARGS}) \mid E.E \quad \text{ARGS} ::= E \mid E, \text{ARGS}$

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1. Surface projection π_Σ

erase names but preserve punctuation

$\pi_\Sigma(\mathbf{x.f(y)}) = \mathbf{ID.ID(ID)}$

checks that the shape belongs to G_0

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names $\rightarrow$ type slots; API names stay

$$\Pi_\Gamma(\mathbf{x.f(y)}) \ni E\langle X \rangle.f(E\langle Y \rangle)$$

contains all and only well-typed terms

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Correctness criterion

For a target language L with surface fragment G' , construct G_Γ satisfying

$$\pi_\Sigma(\sigma) \in \mathcal{L}(G') \iff \Gamma \vdash \Pi_\Gamma(\sigma) \subset \mathcal{L}(G_\Gamma)$$

for all and only source statements σ accepted by a reference compiler for L .

Grammar modularization

C++ context

```
1 // Filetype: *.cpp
2 #include <string>
3 #include <vector>
4 #include "org/fizz.hpp"
5 #include "com/buzz.hpp"
6 using std::string;
7 // ...
8 int fizz() { return 3; }
9 string buzz(string s) { return s; }
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11 int main(int argc, char* args[]) {
12     string bang = "...";
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```

Grammar contribution

$G_{C++} \cup$

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 $\dots$
 $\{ \text{Int} \rightarrow \text{fizz()}, S \rightarrow \text{fizz}(); \} \cup$
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Nominal typing judgements

Context

Γ contains all declarations visible at the cursor

$\Gamma \vdash e : T$

C++ accepts e and gives it the concrete type T .

C++ program

```
struct Meter {  
    int read(int n) { return n + 1; }  
};  
int main() {  
    Meter m;  
    int n = 2;  
    int x = m.read(n);  
    return x + x;  
}
```

Embedding

$\Gamma \vdash m.read(n) : int$

$\Downarrow$

$E\langle int \rangle \rightarrow E\langle Meter \rangle.read(E\langle int \rangle)$

$\Gamma \vdash x + x : int$

$\Downarrow$

$E\langle int \rangle \rightarrow E\langle int \rangle + E\langle int \rangle$

Embedding Substitute locally bound names with typed slots.

Subtyping: tracking inheritance

Derived-to-base call

`Dog <: Animal`

`Dog` publicly inherits `Animal`, so `d` may be passed to `age` as an `Animal`.

C++ program

```
struct Animal {};  
struct Dog : Animal {};  
int age(Animal) { return 7; }  
int main() {  
    Dog d;  
    int x = age(d);  
    return x;  
}
```

Embedding

$$\Gamma \vdash \text{age} : \text{Animal} \rightarrow \text{int}$$
$$\Downarrow$$
$$E\langle \text{int} \rangle \rightarrow \text{age}(E\langle \text{Dog} \rangle)$$
$$E\langle \text{int} \rangle \rightarrow \text{age}(E\langle \text{Animal} \rangle)$$

`age` is contravariant in the input.

Keep the check local

Only track well-scoped subtypes and embed each subtype monomorphically.

Templates: try concrete types, keep valid rules

C++ program

```
#include <string>
#include <vector>
using std::string;
template<class T>
T twice(T x) { return x + x; }
int main() {
    int i = 2; int n = twice(i);
    string h = "ha"; string s = twice(h);
    return n == 4 || s == "haha";
}
```

Try three concrete types

int ✓ accepted

std::string ✓ accepted

std::vector<int> ✗ rejected

The vector case fails because $x + x$ is not defined for vectors.

Keep the successful rules

$$\Gamma \vdash \text{twice}(i) : \text{int} \implies E(\text{int}) \rightarrow \text{twice}(E(\text{int}))$$
$$\Gamma \vdash \text{twice}(h) : \text{std::string} \implies E(\text{std::string}) \rightarrow \text{twice}(E(\text{std::string}))$$

Sound: every kept rule compiles.

Complete within this set: all three were tried.

Class templates: invariant by default

Subtyping does not lift automatically

`Dog <: Animal` $\nRightarrow$ `F<Dog> <: F<Animal>`

Class templates may optionally have a constrained converting constructor.

API-provided covariance

```
std::shared_ptr<Dog> pd;  
std::shared_ptr<Animal> pa = pd;  
// accepted
```

`shared_ptr` explicitly supplies the conversion.

Invariant specialization

```
std::vector<Dog*> ds;  
std::vector<Animal*> as = ds;  
// rejected
```

The two `vector` specializations are unrelated.

CFG consequence

enumerate ground states; validate exact contexts

Never infer that `F<D>` converts to `F<B>` merely from `D <: B`. Emit a rule only when the C++ API supplies and accepts that conversion.

Results: Size of generated CFGs

Evaluation